LAB MANUAL – 2D ARRAYS (C++)

Programming Fundamentals Lab OBE-Aligned Document

1. Lab Title

Working with Two-Dimensional (2D) Arrays in C++

2. Lab Purpose

This lab introduces students to 2D arrays, their declaration, initialization, input, display, and processing using nested loops. Students will perform tasks mapped to CLOs following the OBE approach.

3. OBE Statement

This lab is designed according to Outcome-Based Education (OBE) principles. Each activity supports measurable CLOs.

4. CLO Mapping

CLO 1 – Understand 2D array structure (C2)

CLO 2 – Apply nested loops for 2D arrays

(C3) CLO 3 – Analyze data using 2D arrays

(C4)

5. Learning Objectives

1. Declare and initialize 2D arrays.

2. Store and retrieve data using nested loops.

3. Process 2D array data.

4. Model real-world data using 2D arrays.

6. Lab Instructions

Follow proper indentation, use meaningful variable names, comment code, test with different

inputs, and name files properly.

7. **Example Program**

```cpp
#include <iostream>
using namespace std;

int main() {
    int arr[2][3];

    cout << "Enter 6 values:\n";
    for(int i = 0; i < 2; i++) {
        for(int j = 0; j < 3; j++) {
            cin >> arr[i][j];
        }
    }

    cout << "\nYou entered:\n";
    for(int i = 0; i < 2; i++) {
        for(int j = 0; j < 3; j++) {
            cout << arr[i][j] << " ";
        }
        cout << endl;
    }

    return 0;
}
```

8. **Lab Activities**

Activity 1 (CLO 1 – C2: Understand)

Write a program to declare a 3×3 integer array and initialize it with your own values. Print the array in matrix form.
OBE Goal: Students will understand how 2D arrays store data in rows and columns.

Activity 2 (CLO 2 – C3: Apply)

Write a C++ program to input values in a 2×4 array from the user and display them in table format.
OBE Goal: Students will practice input/output operations with multidimensional arrays.

Activity 3 (CLO 2 – C3: Apply)

Create a program that stores a **restaurant menu** in a 2D array where rows = items, columns = sizes. Display the menu.
OBE Goal: Students will connect code concepts with real-life applications.

Activity 4 (CLO 3 – C4: Analyze)

Write a program to calculate the **row-wise sum** of a 3×3 matrix.
OBE Goal: Students will analyze data stored in arrays using nested loops.

Activity 5 (CLO 3 – C4: Analyze)

Write a program to find the **maximum value** in a 4×4 array.
OBE Goal: Students will learn to search and analyze data patterns.

Activity 6 (CLO 2 – C3: Apply)

Write a program to count how many **even and odd numbers** are stored in a 2D array entered by the user.
OBE Goal: Students will apply control structures with multidimensional arrays.

Activity 7 (CLO 3 – C4: Analyze)

Write a program that multiplies **two 2×2 matrices** and displays the resulting matrix.
OBE Goal: Students will perform operations that require step-by-step analysis.

Activity 8 (CLO 1 – C2: Understand)

Create a program to display the **diagonal elements** of a 3×3 matrix.
OBE Goal: Students will understand index-based access in arrays.

Activity 9 (CLO 3 – C4: Analyze)

Write a program that searches for a specific number in a 2D array.
Print whether it is found and its row/column location.
OBE Goal: Students will analyze array data using nested iterations.

Activity 10 (CLO 2 – C3: Apply)

Write a program that stores temperature readings of 7 days and 3 times per day (morning, afternoon, night) in a 2D array. Display the table.
OBE Goal: Students will model real-world datasets using 2D arrays.

9. **Expected Outcomes**

Students can use 2D arrays, nested loops, and perform computations.